

# Phase 3: Validation Report

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## 1 Introduction

This report documents the validation, performance benchmarking, and experimentation phase of the Boids Interactive Simulation project. The primary goal is to verify that the implemented steering behaviors correctly produce emergent flocking patterns, evaluate the computational efficiency of spatial optimization strategies, and analyze the impact of environmental parameters on prey survival.

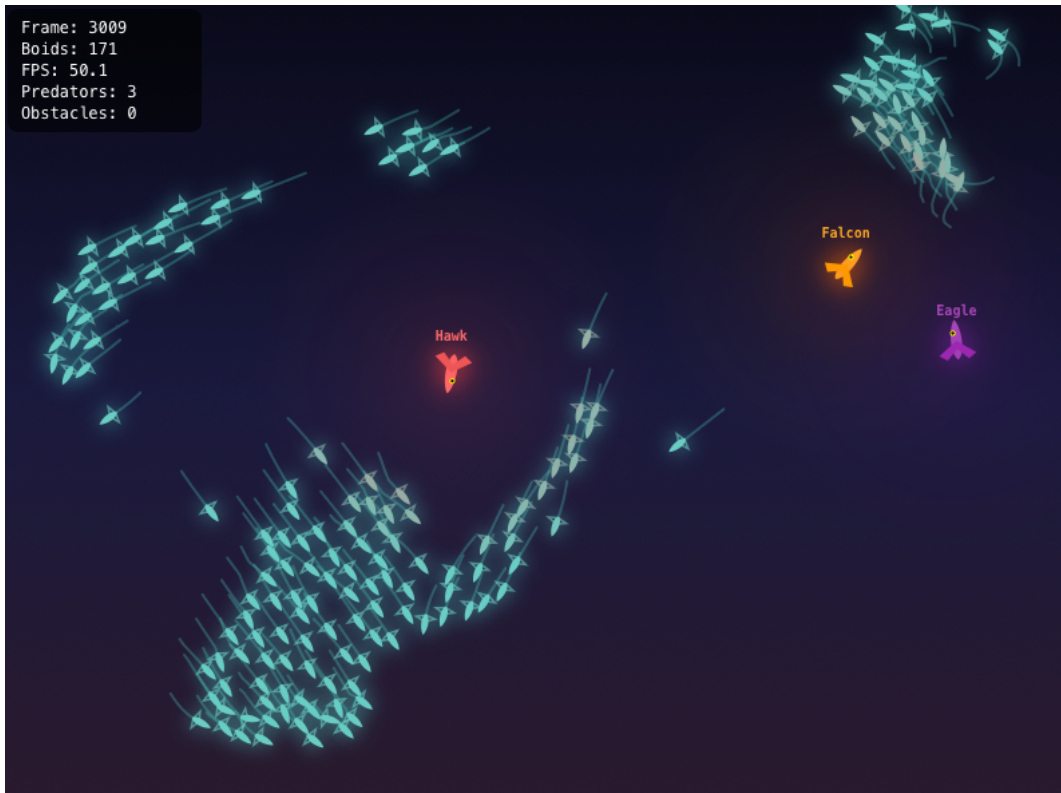


Figure 1: Live simulation demonstrating emergent flocking behavior with 171 boids and 3 predators. The flock exhibits natural splitting behavior in response to predator positioning.

## 2 Behavioral Validation and Speed Dynamics

The simulation was subjected to rigorous testing to ensure adherence to the three fundamental rules: separation, alignment, and cohesion. Initializing from random positions, agents consistently form cohesive flocks within thirty seconds. In the final implementation, boids maintain consistent spacing without overlapping, a result achieved by refining the protected range and separation strength. Within established flocks, boids exhibit high velocity correlation, where direction changes propagate fluidly through the group.

A critical observation during the testing of the interactive demo is that behavioral realism is significantly influenced by the boids' maximum speed. It was observed that higher speeds generally produce results that appear more "natural" to the human observer compared to lower speed settings. This phenomenon is likely an artifact of the simulation structure; at low speeds, the discrete nature of the time-step updates can make steering adjustments appear stiff or mechanical. At higher speeds, the boids possess greater momentum, allowing steering forces to act as smooth perturbations rather than isolated jerks. This better mimics the inertial flight patterns of biological entities and allows emergent patterns to dominate the visual experience.

Complex emergent behaviors were also verified, including wheeling (coordinated turns at boundaries), splitting around predators, and rejoining after threats pass. The "flash expansion" behavior, a sudden dispersal when a predator enters the flock center, confirms that predator-avoidance logic integrates smoothly with the primary flocking algorithm.

## 3 Parameter Exploration

Systematic parameter variation was conducted to understand how each parameter influences flocking behavior. The following parameters were explored:

### 3.1 Number of Boids

| Count | Behavior          | Observations                                                                   |
|-------|-------------------|--------------------------------------------------------------------------------|
| 30    | Sparse flocking   | Small, tight groups; often forms 2-3 separate flocks that rarely merge         |
| 50    | Moderate flocking | Single cohesive flock typically forms; good balance of density and visibility  |
| 100   | Dense flocking    | Rich emergent patterns; clear internal structure visible within flock          |
| 200   | Complex dynamics  | Multiple sub-flocks; more splitting/merging events; performance remains smooth |

Table 1: Effect of boid count on flocking behavior.

### 3.2 Perception Radius

| Range (px) | Behavior   | Observations                                                         |
|------------|------------|----------------------------------------------------------------------|
| 20         | Local only | Many small, independent clusters; poor global coordination           |
| 50         | Balanced   | Single flock forms reliably; good coordination without over-coupling |
| 100        | Global     | Very tight, uniform flock; loses interesting internal dynamics       |
| 150        | Excessive  | Near-instant alignment; flock behaves almost as rigid body           |

Table 2: Effect of visual range on flock coordination.

### 3.3 Maximum Speed

| Speed | Behavior      | Observations                                                                    |
|-------|---------------|---------------------------------------------------------------------------------|
| 1.5-4 | Mechanical    | unnatural flowing motion; easy to observe individual behaviors                  |
| 4-6.0 | Fast, natural | smooth but harder to follow; boundary turns become smoother                     |
| 8.0   | Very fast     | Requires stronger turn factor to prevent boundary escape and obstacle avoidance |

Table 3: Effect of maximum speed on motion characteristics.

### 3.4 Rule Weights

The relative importance of separation, alignment, and cohesion dramatically affects flock character:

| Configuration                                 | Resulting Behavior                                               |
|-----------------------------------------------|------------------------------------------------------------------|
| High separation (0.3), low cohesion (0.0005)  | Dispersed cloud; boids spread apart and rarely form tight groups |
| Low separation (0.05), high cohesion (0.01)   | Dense clumps; boids cluster tightly, sometimes overlapping       |
| High alignment (0.15), low others             | Parallel streams; boids align but don't cluster spatially        |
| Balanced (sep=0.15, ali=0.08-0.12, coh=0.002) | Natural flocking; recommended default configuration              |

Table 4: Effect of rule weight balance on flock structure.

### 3.5 Parameter Exploration Summary

The most realistic flocking patterns emerge with the following parameter setting (High Speed Default):

| Parameter           | Recommended Value |
|---------------------|-------------------|
| Number of boids     | 75-125            |
| Visual range        | 60 px             |
| Protected range     | 12 px             |
| Maximum speed       | 6.0               |
| Separation strength | 0.15              |
| Alignment factor    | 0.12              |
| Cohesion factor     | 0.002             |
| Turn factor         | 0.2               |
| Boundary margin     | 75 px             |

Table 5: Recommended parameter configuration for realistic flocking.

## 4 Performance Benchmarking

The implementation of a KD-Tree spatial partitioning system was essential for achieving real-time performance. Figure 2 illustrates the performance disparity between the naive  $O(n^2)$  approach and the optimized  $O(n \log n)$  KD-Tree implementation.

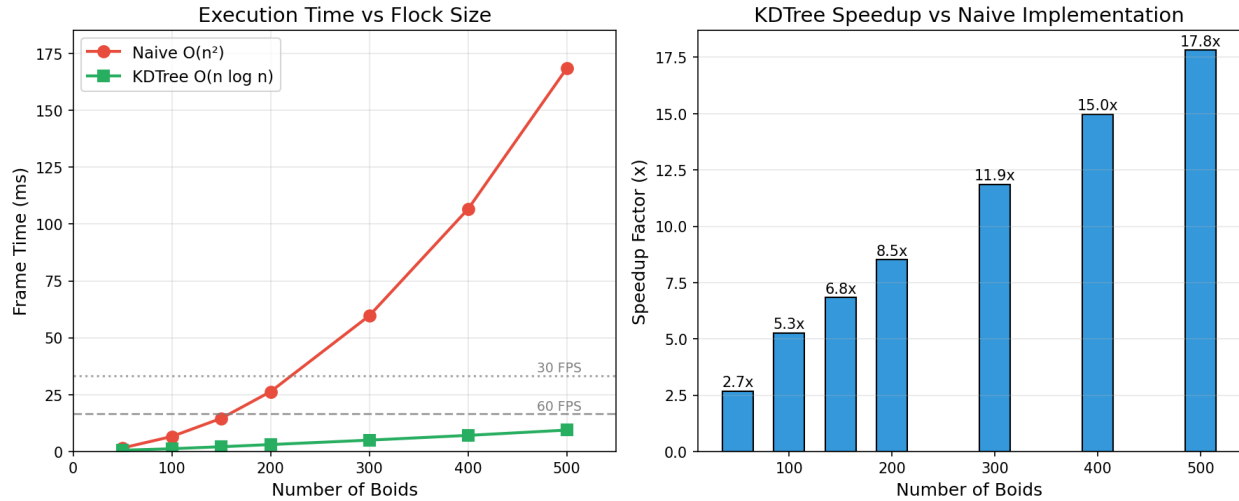


Figure 2: Performance comparison: Execution time vs. Flock Size (left) and Speedup factor (right). The naive implementation exceeds the 60 FPS budget at approximately 150 boids.

The benchmark data reveals that the naive implementation exceeds the 16.67ms frame budget (required for 60 FPS) at approximately 150 boids and crosses the 33.3ms threshold (30 FPS) at 250 boids. In contrast, the KD-Tree maintains performance well below the 60 FPS line even with 500 boids. The speedup factor scales significantly with the number of agents, reaching a 17.8 $\times$  improvement at 500 boids. This optimization is critical for the interactive application, allowing for dense flocks and multiple predators without degradation in user experience.

## 5 Predator-Prey Dynamics Analysis

A systematic parameter sweep of 125 runs was conducted to quantify how predator speed and prey avoidance strength influence survival and flock structure. The results, visualized in Figure 3, indicate that predator speed is the most dominant factor in the system.

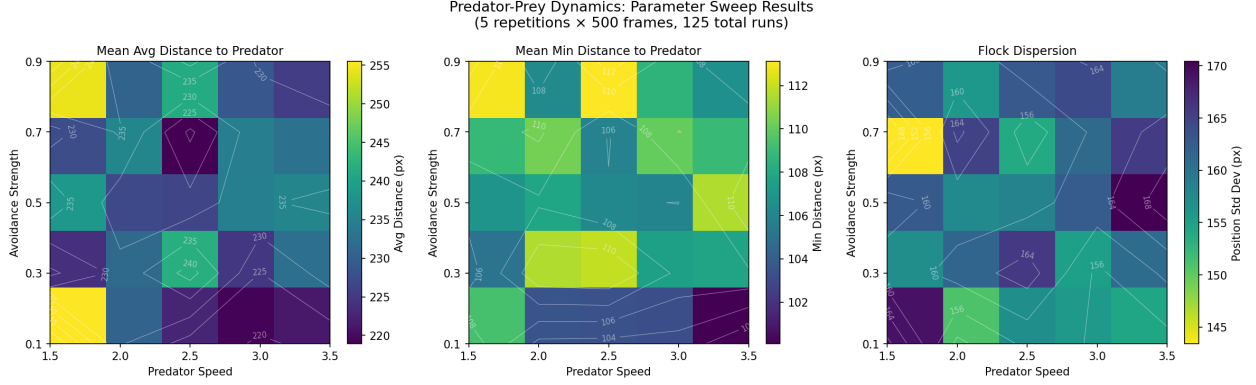


Figure 3: Predator-prey parameter sweep results: Mean average distance (left), mean minimum distance (center), and flock dispersion (right).

The "Mean Min Distance to Predator" heatmap (center panel) shows a clear gradient: as predator speed increases to 3.5, the closest approach distance drops to approximately 100px regardless of the prey's avoidance strength. This suggests that once a predator exceeds a certain speed threshold, prey reaction is physically limited by the simulation's turn radius. Avoidance strength was found to have diminishing returns; increasing it from 0.1 to 0.9 improved minimum approach distance by only about 10%. Furthermore, flock dispersion (right panel) increases under pressure, indicating that high-intensity evasion causes the flock to fragment and scatter, reaching position standard deviations of  $\sim 170$ px.

## 6 Known Limitations

The current implementation has several limitations identified during validation:

- Small obstacle avoidance: Obstacles with radius  $< 20$ px do not trigger sufficiently strong avoidance; boids may clip through at high speeds.
- Screen size dependency: The demo works best with a square or near-square browser window (800×600 to 1000×800 pixels). Full-screen on wide monitors causes layout issues.
- Docker performance: Currently achieves  $\sim 48$ – $50$  FPS versus  $\sim 60$  FPS in local development. Container optimization is planned.
- Predator selection: Predators are added by count (1–5) with automatically assigned strategies. Users cannot currently select specific strategies (e.g., "5 Hawks").

## 7 Reflection and Limitations

The project was developed with GenAI assistance, producing 501 passing tests. Understanding the algorithm and project requirements was useful in identifying implementation issues and visually inspecting each feature before full integration. Personal intervention remained necessary for fine-tuning parameters like the turn and avoidance factors was initially caused unnatural reversals.

Furthermore, developer experience with full-stack projects was valuable in outlining project structure, tool usage and the process of gradual component integration. GenAI (especially Claude AI) has a strong tendency to avoid gradual development along with the user. The power of this model is also its greater weakness in my personal experience. Unclear instructions and prompts can often lead to poorly thought implementations with overcomplicated structures that far exceed the spec's requirements. Although impressive at first glance, the ability of the LLM to generate thousands of lines of code at once has lead me to confusion and undesirable results on a number of occasions. Specifying a clear project structure, strict module and integration testing, and gradual development of the stack led to (hopefully) positive results. Finally, it is worth mentioning that this project was a valuable learning experience for me. While Typescript has never been my greatest strength, I appreciated the opportunity to demystify the full-stack development of an interactive demo and gained valuable insights from the thought and implementation process required.

My goal is to continue working on the project to refine simulation features and current limitations and possibly expand the demo's capabilities to three dimensions. Unfortunately, creating such an extension with Three.js exceeds my level of proficiency at the moment. Combined with additional physics requirements for the flocking rules, I estimated that the extension would not be a viable option within a reasonable time-frame. That being said, I hope to implement a working 3D version in the coming months. (which I would gladly share with you!)